

Ellie Hara, PhD

SRI Postdoctoral Fellow, University of Utah (ellie.hara@utah.edu)

CURRENT ROLE

- Aug 2026 – **SRI Fellow, College of Science, University of Utah**
current
- Jul 2024 – **Postdoctoral Researcher, RARE Center**
May 2026 Advisors: Professor Karyn Rogers, Professor Matthew Pasek

EDUCATION

- 2017 – 2023 **PhD in Geological Sciences, University of Colorado Boulder (Dec 21st 2023)**
Thesis: Cyanide dynamics in serpentinizing systems; implications for prebiotic chemistry
Advisor: Professor Alexis Templeton
NSF Graduate Research Fellowship Program Fellow
- 2014 – 2016 **B.S. in Geological Sciences Cum Laude, University of Southern California**
Thesis: Earthquake Driven Erosion of Organic Carbon
Advisor: Professor A. Joshua West
- 2012 – 2014 **El Camino College & Santa Monica Community College, Non-degree seeking**

PUBLICATIONS

- 2025 Cyanide Storage on Ferroan Brucite ($\text{Mg}_x\text{Fe}_{1-x}(\text{OH})_2$): Implications for Prebiotic Chemistry, *Minerals*
Hara E.K. and Templeton A.S.
- 2024 Glycine synthesis from nitrate and glyoxylate mediated by ferroan brucite: An Integrated pathway for prebiotic amine synthesis, *PNAS*
Chimiak L., **Hara E.**, Sessions, A., Templeton A.S.
- 2024 Releasing Cyanide from Ferrocyanide through Carbon Monoxide Ligand Exchange in Alkaline Aqueous Environments, *ACS Earth and Space Chemistry*
Hara E.K. and Templeton A.S.
- 2023 Detecting organics with DUV Raman and Fluorescence, *Nature Reviews Earth & Environment*, **Hara E.** [Non-peer-reviewed article]

MANUSCRIPTS SUBMITTED OR IN PREP

- In review* Reproducing submarine hydrothermal complexity: A transparent high-pressure flow reactor for mineral growth, aging, and prebiotic chemistry
Boé G., **Hara E.**, Ercicek F., Cario A., Rogers K., Johnson-Finn K., Tassaing T., Marre S.

<i>In review</i>	Amino acid synthesis from pyruvate and nitrate on ferroan brucite: pathways for simultaneous alanine and glutamine formation Chimiak L., Hara E. , Sessions, A., Templeton A.S.
<i>Submitted</i>	Accelerated Carbon Dioxide Drawdown following a Large Earthquake Li G., West J.A., Fischer W., Zhangdong J, Hara E. , Zhang F., Wang J., Hammond D., Berelson W., Qiu H., Densmore A., Lamb M., Hilton R.
<i>In prep</i>	Mineralogy transformations of Mg phosphates and implications for planetary phosphorus availability Hara E. , Rogers K., Pasek M.
<i>In prep</i>	LEGOS: Laboratory Experiments for Gauging Organic Stability- An educational framework for prebiotic chemistry exploration Riggi V., Hara E. , Thompson B., Rogers K.

PROFESSIONAL EXPERIENCE

2026 – current	SRI Postdoctoral Fellow, University of Utah <i>Research and Mentorship:</i> Building an undergraduate research lab and developing research streams to give undergraduate students meaningful research experience and skillsets from their first year.
2024 – 2026	Postdoctoral Researcher, Rensselaer Polytechnic Institute <i>Research:</i> Mineral-facilitated organic chemistry reactions in custom-built hydrothermal flowthrough reactors <i>Instrumentation:</i> H-NMR, ATR-FTIR, Raman, XRD, SEM/EDS, HPLC-UV-MS <i>Mentorship:</i> Mentoring 4 undergraduates and 1 high school student
2017 – 2023	Graduate Student, University of Colorado – Boulder <i>Research:</i> Investigating cyanide interactions in serpentinizing environments through controlled experiments with brucite. Developed methodology for anaerobic, organic clean experimental set up with analytical expertise in gas chromatography, Raman spectroscopy, Infrared spectroscopy, Ion chromatography, and H-NMR. <i>Analytics:</i> R, OriginPro and MestreNova <i>Mentorship:</i> Mentored an undergraduate student in geochemical lab techniques and a broad range of analytical instrumentation
2016 – 2017	Lab Technician, Jet Propulsion Laboratory Characterizing the Raman spectra of crystalline and amorphous histidine in proxy Mars paleolakes to assist with future data interpretation and detection limit determination for the SHERLOC instrument on the Perseverance Rover.
2015 – 2016	Intern, Jet Propulsion Laboratory <i>Advisors:</i> Dr. Kenneth Williford, Dr. Michael Russell, Dr. Rohit Bhartia SEM/EDS, and petrographic microscope analysis to locate kerogen in thin sections of a 3.5-billion-year-old carbonate formation. DUV Raman and Fluorescence mapping to analyze serpentinite samples from Cedars. Trained new users on instrumentation.

- 2014 – 2016 **Undergraduate Student Researcher, University of Southern California**
Investigated earthquake-driven organic carbon transport through sediment analysis post-Wenchuan earthquake. EA-IRMS sample preparation and analysis.
- 2014 **Summer Intern, Center for Dark Energy Biosphere**
EA-IRMS sample preparation and analysis.

PRESENTATIONS

Oral Presentations

- May 2026 *First Friday Astronomy, Boise State University (Invited Speaker - Public Lecture)*
Central Idaho Dark Sky Reserve Lecture, Hailey Public Library
From Stardust to Cells: The Unsolved Mystery of Life's Origin
- Mar 2026 *RPI Department of Earth and Environmental Sciences Seminar*
Mineralogy transformations of Mg phosphates and implications for planetary phosphorus availability
- Jun 2024 *PCE3 (Invited Speaker)*
Releasing cyanide from ferrocyanide in alkaline aqueous systems
Hara, E., Templeton A.S.
- Feb 2024 *CO World Seminar Series (Invited Speaker)*
Cyanide dynamics in serpentinizing systems: implications for prebiotic chemistry
Hara, E., Templeton A.S.
- July 2023 *Goldschmidt*
Amino acid synthesis driven by ferroan brucite-mediated chemistry
Chimiak L. M., **Hara E.**, Ellison E. T., Eiler J., Sessions A., VanderVelde D., Templeton A S.
- May 2022 *AbSciCon*
Releasing cyanide from ferrocyanide in alkaline aqueous systems, AbSciCon
Hara, E., Templeton A.S.
- Sept 2021 *Rocky Mountain Geobiology Symposium*
Releasing cyanide from ferrocyanide by carbon monoxide ligand exchange in alkaline and hyperalkaline aqueous environments
Hara, E. K., Templeton A. S.
- Apr 2017 *AbSciCon*
Detection of Subsurface Organics using Deep UV Raman and Fluorescence Spectroscopy: Implications for SHERLOC on Mars 2020
Carrier B. L., Beegle L. W., Bhartia R., Abbey W. J., **Hara E. K.**
- A Terrestrial Analogs Database for Mars and Europa Mission Operations Support
Doloboff I. J., Eshelman E. J., Wagner G., **Hara E.**, Paez V. M., Orphan V. J., Amend J. P., Beegle L. W., Bhartia R.

Poster Presentations

- Jan 2024 *Joint CO World & 12th ELSI Symposium*
Releasing cyanide from ferrocyanide by carbon monoxide ligand exchange in alkaline and hyperalkaline aqueous environments
Hara, E., Templeton A.S.
- May 2023 *AbGradCon*
Releasing cyanide from ferrocyanide by carbon monoxide ligand exchange in alkaline and hyperalkaline aqueous environments
Hara E., Templeton A. S.
- Apr 2023 *Rocky Mountain Geobiology Symposium*
Iron-nickel brucite cannot facilitate cyanide formation but may be able to facilitate methane oxidation to methanol, *Rocky Mountain Geobiology Symposium*
Hara, E., Chimiak L., Templeton A.S.
- Dec 2017 *AGU Annual Meeting*
A SHERLOC Study: Detection of Organics in Simulated Martian Soil using Deep UV Raman Spectroscopy
Abbey W., Bhartia R., Carrier B., Doloboff I., **Hara E.**, Beegle L.
- Apr 2017 *AbSciCon*
Analogue Sample Development for Simulation of Mars Paleolake Environments: Assessing Spectral Effects
Hara, E.K., Bhartia R., Abbey W.J., Carrier B.L., Wagner G., Beegle L.W.
- May 2017 *Lunar and Planetary Science XLVIII*
Multi-Instrument Database (MIND): Cloud-Based Database System for DUV Raman and Fluorescence Research and Flight Instrument Engineering Telemetry
Doloboff, I. J., Paez V. M., Eshelman E. J., **Hara E.**, Wagner G., Orphan V. J., Amend J. P., Beegle L. W., Bhartia R.
- Dec 2016 *AGU Annual Meeting*
Earthquake-driven erosion of organic carbon at the eastern margin of the Tibetan Plateau
Li, G., **Hara E. K.**, Hammond D. E., Hilton R., G.
- Mar 2016 *Southern California Geobiology Symposium*
Earthquake Driven Erosion of Organic Carbon, *SoCal Geobiology Symposium*
Hara, E.K., Li, G., West J.A., Hammond D., Berelson, W.
- Apr 2015 *Southern California Geobiology Symposium*
Characterization of organic carbon transport in the aftermath of a large earthquake, *USC*
Hara, E.K., Li, G., West J.A.
- Apr 2015 *Undergraduate Symposium for Scholarly and Creative Work*
Characterization of organic carbon transport in the aftermath of a large earthquake
Hara, E.K., Li, G., West J.A.
1st place in Physical Sciences and Engineering

FUNDING AND FELLOWSHIPS

2025	ASU NASA SCoPE Workshop Travel Grant (\$1200)
2024	ASU SciAct Affiliate AGU Travel Grant (\$2200)
2022	Department Travel Grant, CU Boulder Department of Geological Sciences (\$1200)
2022	University of Colorado Boulder Graduate Student Travel Grant (\$500)
2017 – 2022	NSF Graduate Research Fellowship Program (GRFP) Fellow (\$138,000 stipend + tuition)
2015, 2016	Student Opportunities for Academic Research (SOAR) Grant (\$1000, \$1000)
2015	WiSE Research Fellowship, USC (\$500)

AWARDS AND HONORS

2026	MSCA Postdoctoral Fellowship Seal of Excellence
2021	International Association for Geoscience Diversity, Student Pathways Scholarship (\$500)
2016	Estwing Hammer Award, USC Department of Earth Sciences (2016)
2015 – 2016	Jessie Ingram Endowed Scholarship, USC
2015	1 st place in Physical Sciences, Mathematics and Engineering, Undergraduate Symposium for Scholarly and Creative Work, USC
2015	Richard O. Stone Scholarship, USC

TEACHING EXPERIENCE AND MENTORING

Teaching Experience and Coursework

Fall 2026	Instructor of Record, University of Utah SCI 1500: Principles of Scientific Inquiry
Spring 2025	Guest Lecture, Rensselaer Polytechnic Institute Introduction to Astrobiology
Spring 2024	Adjunct Faculty, University of Colorado – Boulder GEOL 1020: History of a Habitable Planet (159 students)
Spring 2024	Course Assistant, University of Colorado – Boulder Assistant for GEOL 1030 Mentored new graduate student teaching assistants Assisted with adapting field labs into accessible virtual lab modules
2022 – 2023	Lead Teaching Assistant, University of Colorado – Boulder Led teaching assistant training for GEOL 1030: Introduction to Geology Laboratory Maintained course infrastructure (Canvas), managed department tutoring and served as liaison between faculty and admin
2019, 2020	Pedagogical Coursework, University of Colorado – Boulder Fall 2019: Communicating Earth Science to the Public Fall 2020: Teaching and Learning in Post-Secondary Education
2017 – 2018	Teaching Assistant, University of Colorado – Boulder GEOL 2001 (Planet Earth), GEOL 3320 (Introduction to Geochemistry), GEOL 4716 (Environmental Field Geochemistry)

Mentoring

- 2025 – present **LEGOS Lead Analytical Mentor, RARE Center**
Trained and supervised 1 high school and 4 undergraduate students in geochemistry laboratory techniques and data analysis using Excel and R. The high school student is a 2026 Regeneron International Science and Engineering Fair Finalist
- 2020 – 2022 **Research Mentor, University of Colorado – Boulder**
Mentored a student in geochemical wet laboratory and analysis techniques
Student is now an environmental geochemist

OUTREACH AND SERVICE

Professional Service

- 2026 Primary Session Convener, AbSciCon
- 2025 – present Early Career Committee, NASA RCN PCE3
- 2025 Secretary, NASA Exobiology Review Panel
- 2024 – present Subject Matter Expert, NASA Community College Network
- 2024 – present AGU SciAct Affiliate, NASA SCoPE
- 2020 – 2022 Lead Organizer, Graduate Admissions to the Earth Sciences (GATES)
- 2020 – 2021 Co-founder and Coordinator, Geological Sciences Graduate Association (GSGA)
- 2019 – 2020 Faculty Meeting Graduate Representative

Reviewer for: Geochimica et Cosmochimica Acta, ACS Omega, NASA Exobiology External Reviewer

Outreach

- 2025 Guest Lecture, NASA Community College Network Seminar
- 2024 – present Letters to a Pre-scientist
- 2022 – present Mentor, Geosciences Education & Mentorship Support (GEMS)
- 2020 – 2022 Peer Mentor, University of Colorado – Boulder
- 2017 Primary Science Instructor, World Space Expedition Program, Madrid Middle School
- 2016 Coursework development, EcoVoices Education

Additional outreach includes participation in Skype a Scientist, Interview with a Scientist, Letters to a Pre-Scientist and various panels at schools and community colleges.

PROFESSIONAL DEVELOPMENT

- 2025 ASU NASA SCoPE Science Communication Workshop
- 2024 AGU/SciAct Affiliate Science Communication Workshop
- 2022 AbGradCon Proposal Writing Workshop

PROFESSIONAL AFFILIATIONS

PCE3, Early Career Committee
Future Leaders of Ocean Worlds (FLOW)
American Geophysical Union
Phi Beta Kappa